

Roll No.....

MID SEMESTER EXAMINATION SEPTEMBER - 2025

Name of the Program: B. Tech.

Semester: I

Name of the Course: Engineering Physics

Course Code: TPH 101

Time: 90 Minutes

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions.

(ii) Each question carries 10 marks

Q1	(10 marks)	CO 1
(a)	Deduce the necessary expression to determine the refractive index of unknown transparent liquid using Newton's ring experiment.	
OR		
(b)	Define the phenomenon of interference and derive the expression for the resultant intensity. Also, write the conditions for maximum and minimum intensities.	
Q2	(10 marks)	CO 1
(a)	Discuss the method to determine the wavelength of light using the Fresnel Biprism experiment.	
OR		
(b)	(i) Deduce the expression for fringe width in a wedge-shaped film. (ii) In the Fresnel Biprism experiment, the angle of the biprism is $\frac{1}{2}$ degree and the refractive index is 1.5. The distance between the source and the biprism is 30cm. When a source of wavelength $5893 \text{ \AA}$ is used, interference fringes are obtained on the screen; the distance between the biprism and the screen is 60cm. Calculate the fringe width and maximum number of fringes.	
Q3	(10 marks)	CO 1
(a)	In single slit Fraunhofer diffraction, derive the expression for resultant intensity. Also, find the angular separation between the first and second minima.	
OR		
(b)	(i) Define the resolving power of a grating; derive the required expression. (ii) In a grating at a 30 degree angle of diffraction, the spectral line of $5000\text{\AA}$ in a certain order is superimposed on the line of $4000\text{\AA}$ of the next higher order. Find the lines/cm in a grating.	
Q4	(10 marks)	CO 2
(a)	Write the working principle of an optical fiber, also derive the expression for acceptance angle and numerical aperture.	
OR		
(b)	Calculate the refractive index of the core and cladding material of a fiber having Numerical Aperture = 0.25 and fractional refractive index change $(\Delta\mu_r) = 0.012$.	
Q5	(10 marks)	CO 2
(a)	Write short notes on different types of optical fiber and their applications. Describe how the conditions of total internal reflection are implemented in an optical fiber.	
OR		
(b)	Find the fractional refractive index and numerical aperture for an optical fiber with refractive indices of core and cladding as 1.5 and 1.45 respectively.	

